Zachary Lieberman

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SUMMARY

Software engineer with 3 years of experience building backend services, data platforms, and internal tools at enterprise scale. Skilled in Python, TypeScript, SQL, and cloud-native architecture on AWS. Track record of designing systems handling 200M+ daily executions, reducing operational costs by over $1M, and scaling platforms across multiple regions.

SKILLS

Languages & Frameworks: Python (Flask, FastAPI, boto3, SQLAlchemy, Alembic, Pydantic), TypeScript, Java, SQL, Go, Bash, REST APIs, JSON, YAML

Frontend: React, Vite, React Router, Tailwind CSS, D3.js

Data & APIs: GraphQL, ClickHouse, PostgreSQL (asyncpg), Aurora, RDS, DynamoDB, MySQL, Anthropic Claude API, GitHub API

Cloud (AWS): Lambda, ECS/Fargate, EC2, ASG, AMI, S3, SQS, API Gateway, CloudFormation, SAM, RDS, Batch, VPC, IAM, ECR

DevOps & Testing: Docker, Docker Compose, Kubernetes, Terraform, Jenkins, Git, GitHub Actions, HashiCorp Vault, JFrog, CI/CD, pytest, Vitest, ruff, ESLint, Cloud Custodian, n8n

EXPERIENCE

SpaceX, Build Reliability Engineering (BRE), Software Engineer Starting Nov 2026

Hawthorne, CA

- Software Engineer on the BRE Process Engineering team.

Capital One, Slingshot Product Team, Software Engineer Dec 2025 - May 2026

Los Angeles, CA

- Built end-to-end dashboard features (Warehouse Performance, Contract Analysis, Data Explorer) giving customers real-time operational visibility into Snowflake cost and usage data.
- Developed interactive UI components, including combo-box filters and bar-chart slicing, letting users drill into metrics without leaving the dashboard.
- Wrote ClickHouse SQL queries powering analytics across large-scale Snowflake datasets, optimizing for performance at high query volume.
- Updated TypeScript backend to support GraphQL queries, enabling flexible, efficient data fetching for dashboard features.

Capital One, Cloud Custodian Platform Team, Software Engineer Aug 2023 - Dec 2025

McLean, VA

- Owned and evolved an enterprise cloud-compliance platform serving 3,000+ AWS accounts with 200M+ daily policy evaluations.
- Created a resource-aware batch evaluation system (Python, Lambda, SQS, DynamoDB) that skips empty accounts, saving over $1M annually in compute costs.
- Scaled the platform from 2 to 6 AWS regions (300% increase in daily executions) while maintaining performance and compliance coverage.
- Led migration of 100+ services to a new account model, configuring VPCs, security groups, and CloudFormation templates with zero downtime.
- Built a policy dry-run system (PostgreSQL, ECS) that previews the impact of 10,000+ compliance policies before execution, eliminating unintended resource actions.

PROJECTS

Job Application Assistant, Full-Stack AI App (FastAPI, React/TypeScript, Claude API) 2026

- Built a full-stack application (async FastAPI + SQLAlchemy/PostgreSQL backend, React/TypeScript/Vite frontend) that integrates the Anthropic Claude API to tailor resumes, generate cover letters, and produce interview prep.

LEADERSHIP

- Mentored 4 interns through Capital One's Technology Internship Program (2024): technical guidance, code reviews, career coaching.
- TechX 2023 Q3 Finalist for AWS Graviton migration (saving $1M+ annually).

EDUCATION & CERTIFICATIONS

University of Virginia, School of Engineering and Applied Sciences, B.S. Computer Science

AWS Certified Solutions Architect, Associate (2023)